

MATH8009 – Regression Tutorial & Lab Exercise Sheet 2

1) Import the Edu_salary.csv dataset into R; this dataset contains three variables:

- i. Salary (Teachers salary in €000's)
- ii. Years (Years of service)
- iii. Rating (Teaching Performance rating)

- (a) Create scatter plots the data for 1) Salary versus Years and 2) Salary versus Rating in separate plots. Use appropriate labels and units of measurement (where applicable). Describe the relationship between the variables in each case.
- (b) Create a linear model for each case in (a). Write down the regression equations for each model. Show the corresponding regression line in each scatter plot.
- (c) Which model do you think is more accurate? Use appropriate numerical measures to explain your answer.
- (d) Use each model to predict the salary of a teacher with 10 years' service and a rating of 50.
- (e) Create a multiple linear regression model where both Years and Rating are used as predictor variables.
 - a. How does this model compare to the two from part (b)?
 - b. Use this model to predict the salary of a teacher with 10 years' service and a rating of 50.

2) Import the typepressure.csv data into R; this data set contains two variables

- i. Type pressure in PSI
- ii. Mpg (Miles per gallon)

- (a) What are the independent and dependent variables? Describe what relationship you are investigating in this question.
- (b) Create an appropriate scatter plot. Describe the relationship between the variables.
- (c) Create a linear model for this data.
 - a. Show the regression line and data in the same plot.
 - b. Using this linear model, predict what the mpg would be for a tyre pressure of 32 PSI.
 - c. For each of the mpg values calculate the corresponding residuals.

(d) Discuss the model and its fit using the following measures. State the value of each measure and explain what it means.

- a. Correlation coefficient
- b. R-squared
- c. Test statistics
- d. P-values

(e) *[Advanced]* What do you notice about the plot from (b)? Does the data follow a linear pattern? How would you alter the model?

- a. Try fitting a quadratic model.
- b. Plot this model along with the data.
- c. Is this quadratic model an improvement on the linear model? Provide evidence to support your answer.
- d. Use this non-linear model to predict the mpg for a tyre pressure of 32 PSI.

3) *[Advanced]* Import the SBP.csv data into R. This data contains 4 variables

- i. SBP (Systolic Blood Pressure)
- ii. Age
- iii. Weight
- iv. Gender

(a) Identify the type of each of the variables above (numerical or categorical). Recode any categorical variables so that they are binary variables. (Note, you may need to install the package dplyr.)

(b) Create scatter plots of SBP v Age and SPB v Weight. Create linear models of SPB v Age and SPB v Weight. Show the regression lines in each scatter plot. Discuss how good each model is. Use each model to predict the SBP for

- i. A weight of 190 lbs
- ii. An age of 80 years

(c) Investigate the effect of the Gender variable on the data. Does it have a significant relationship with the SBP?

(d) Now fit a model with SPB as the dependent variable and all other variables as independent (predictor) variables. Is the model significant? Can the model be improved?

(e) Use this model to predict the SPB for a female of age 80 years with weight 190 lbs. What is the prediction for a male of the same age and weight?